

State-of-the-Art Literature Review

Neural Field Turing Machines for PDE Solving

Executive Summary

Total Papers Reviewed: 25

Highly Useful: 15 papers

Moderately Useful: 9 papers

Not Useful: 1 paper

Top 5 Priority Papers

1. Recurrent Neural Operators (RNO) - immediate training improvement
2. Active Learning for Neural PDE Solvers - data efficiency for 2D extension
3. Transolver - spatial attention enhancement
4. PhysicsCorrect - training-free correction
5. PDE-Refiner - shock refinement post-processing

Papers from First Analysis

1. Weak Baselines and Reporting Biases

Abstract

Critically examines ML benchmarking practices for fluid PDEs, revealing that weak baselines and selective reporting inflate performance claims, and provides rigorous evaluation protocols.

Usefulness

Rating: **Highly Useful**

Adopt their rigorous metrics and baseline comparisons to ensure your TCAN/Transformer results aren't overoptimistic, especially when claiming generalization across viscosities.

Download

<https://arxiv.org/pdf/2407.07218>

2. Augmenting PINN for 2D Burgers Equation

Abstract

Demonstrates hybrid PINN approach for 2D Burgers by combining physics-informed loss with strategically selected solution data points to overcome spectral bias and improve shock capture.

Usefulness

Rating: Moderately Useful

Directly relevant to your planned 2D Burgers extension (Section 5.1.4); the hybrid data-physics approach could complement your pure data-driven method.

Download

<https://arxiv.org/pdf/2301.07824>

3. Deep Branching Solver for Nonlinear PDEs

Abstract

Proposes branching neural network architecture leveraging probabilistic representations and backward SDEs to solve high-dimensional nonlinear PDEs without requiring spatial discretization.

Usefulness

Rating: Not Useful

Focuses on high-dimensional elliptic/parabolic PDEs using fundamentally different mathematical framework (backward SDEs); doesn't apply to your time-evolution autoregressive approach.

Download

<https://arxiv.org/pdf/2203.03234>

4. Multi-Resolution PDE Learning Framework

Abstract

Introduces multi-resolution learning framework that preserves PDE structure across scales using wavelet decomposition and resolution-adaptive training for spatiotemporal systems.

Usefulness

Rating: Moderately Useful

The multi-scale approach could address your shock formation challenges, though integrating wavelet decomposition would require substantial TCAN redesign.

Download

<https://arxiv.org/pdf/2205.03990>

5. Physics-Informed Neural Operator

Abstract

Combines neural operator framework (learning mappings between function spaces) with physics-informed constraints, enabling parameter-to-solution maps that respect governing equations.

Usefulness

Rating: Highly Useful

Directly relevant to your goal of generalizing across viscosity values without explicit conditioning; could improve your cross-viscosity generalization.

Download

<https://arxiv.org/pdf/2111.03794>

6. Multi-Fidelity Laplace Neural Operators

Abstract

Develops Laplace Neural Operators for parametric PDEs with multi-fidelity training data and built-in uncertainty quantification through Bayesian inference.

Usefulness

Rating: Moderately Useful

Uncertainty quantification addresses a gap in your work, though Laplace approximation adds computational overhead conflicting with training efficiency goals.

Download

<https://arxiv.org/pdf/2502.00550>

7. FEM-Based Physics-Informed Operator Learning

Abstract

Proposes FEM-based neural operator framework that handles irregular geometries and complex boundaries by encoding finite element basis functions into the architecture.

Usefulness

Rating: Highly Useful

Directly addresses your boundary condition issues and will be critical when you introduce geometric obstacles in 2D Burgers.

Download

<https://arxiv.org/pdf/2405.12465>

8. Sparse λ -FNO for Transfer Learning

Abstract

Introduces sparse Fourier Neural Operator variant enabling transfer learning across PDE families and parameter regimes with reduced computational cost through learned sparsity patterns.

Usefulness

Rating: Highly Useful

Addresses your training efficiency concerns and viscosity generalization goals; sparse architectures could accelerate hyperparameter sweeps.

Download

<https://arxiv.org/pdf/2512.17969>

9. Multi-Objective Loss Balancing

Abstract

Presents adaptive loss balancing algorithms that automatically weight competing physics-informed loss terms to prevent gradient pathologies and improve convergence.

Usefulness

Rating: Highly Useful

You use multi-term physics-informed loss; their adaptive weighting could significantly improve training stability and convergence speed.

Download

<https://arxiv.org/pdf/2110.09813>

10. Hybrid Boundary PINNs for Navier-Stokes

Abstract

Develops specialized PINN architecture for Navier-Stokes with complex boundaries by decomposing solution into boundary-fitted and interior components handled by separate networks.

Usefulness

Rating: Highly Useful

Directly applicable to your Navier-Stokes extension and obstacle-laden 2D Burgers; offers concrete solution to padding artifacts.

Download

<https://arxiv.org/pdf/2507.17535>

Papers from Second Analysis (2024-2025)

11. Latent Neural PDE Solver (LNS)

Abstract

Proposes learning PDE dynamics in a compressed latent space using autoencoder projection followed by temporal modeling, reducing computational cost by ~50% while maintaining accuracy.

Usefulness

Rating: Highly Useful

Directly addresses training efficiency; latent-space approach could dramatically speed up TCAN training by operating on coarser representations.

Download

<https://arxiv.org/pdf/2402.17853>

12. Transolver: Fast Transformer for General Geometries

Abstract

Introduces Physics-Attention mechanism that groups mesh points into learned physical states rather than applying attention directly to spatial points, achieving linear complexity.

Usefulness

Rating: Highly Useful

Physics-Attention could be adapted to your spatial dimension, potentially improving shock formation and gradient structure capture.

Download

<https://arxiv.org/pdf/2402.02366>

13. Unisolver: Universal Neural PDE Solver

Abstract

Pre-trains Transformers on diverse PDEs conditioned on equation components (coefficients, boundary conditions, symbols), achieving cross-PDE generalization through explicit physics conditioning.

Usefulness

Rating: Highly Useful

Their PDE-conditioning framework could enable extension to 2D Burgers and Navier-Stokes with minimal architecture changes.

Download

<https://arxiv.org/pdf/2405.17527>

14. DiffusionPDE: Generative PDE Under Partial Observation

Abstract

Uses diffusion models to jointly learn distribution of PDE coefficients and solutions, enabling reconstruction from sparse sensors by treating PDE-solving as conditional generation.

Usefulness

Rating: Moderately Useful

Enables robustness testing and data-efficient training from sparse observations, though requires substantial architectural changes.

Download

<https://arxiv.org/pdf/2406.17763>

15. Physics-Informed Diffusion Models

Abstract

Embeds PDE residuals directly into diffusion model training loss, reducing residual error by 2 orders of magnitude while providing natural regularization against overfitting.

Usefulness

Rating: Moderately Useful

Offers alternative physics-informed loss structure; could inspire hybrid training strategies combining data-driven and physics-informed objectives.

Download

<https://arxiv.org/pdf/2403.14404>

16. Recurrent Neural Operators (RNO)

Abstract

Trains neural operators autoregressively during training to expose model to its own prediction errors, achieving dramatically improved long-rollout stability by eliminating train-test mismatch.

Usefulness

Rating: Highly Useful

Directly addresses error accumulation; full autoregressive training could stabilize your 256-step rollouts and improve cross-viscosity generalization.

Download

<https://arxiv.org/pdf/2505.20721>

17. Active Learning for Neural PDE Solvers

Abstract

Benchmark framework demonstrating active learning reduces average error by 71% vs random sampling by intelligently selecting which PDE parameters and initial conditions to query.

Usefulness

Rating: Highly Useful

Could reduce data requirements while maintaining generalization, critical when extending to expensive 2D Navier-Stokes simulations.

Download

<https://arxiv.org/pdf/2408.01536>

18. PDE-Refiner: Accurate Long Rollouts

Abstract

Iteratively refines one-step predictions using denoising-inspired process to recover high-frequency information lost in standard MSE training, achieving 5-10x longer stable rollouts.

Usefulness

Rating: Highly Useful

Multi-step refinement could be integrated as post-processing to recover shock sharpness lost during autoregressive rollout.

Download

<https://arxiv.org/pdf/2308.05732>

19. WF-PINNs for Burgers Equation

Abstract

Uses weak-form (integral) PDE loss with entropy conditions to stabilize shock capture in Burgers equation, outperforming strong-form PINNs in steep-gradient regions.

Usefulness

Rating: Moderately Useful

Weak-form loss and entropy enforcement could be incorporated as regularization terms, improving shock resolution.

Download

<https://www.nature.com/articles/s41598-025-24427-4> (may require institutional access)

20. APEBench: Autoregressive PDE Benchmark

Abstract

Comprehensive JAX-based benchmark with 46+ PDEs across 1D/2D/3D, differentiable physics for solver-in-the-loop training, and standardized rollout metrics.

Usefulness

Rating: Highly Useful

Provides rigorous evaluation framework; enables fair comparison with state-of-the-art and validates cross-viscosity generalization claims.

Download

<https://arxiv.org/pdf/2411.00180>

21. Latent Neural Operator (LNO)

Abstract

Solves PDEs in latent space with single transformation, reducing parameters by 30% and memory by 50% compared to Transolver while maintaining competitive accuracy.

Usefulness

Rating: Highly Useful

Efficiency improvements directly address your training speed concerns; single latent transformation is simpler than multi-layer approaches.

Download

[NeurIPS 2024 Proceedings](#)

22. Transolver++: Million-Scale Geometries

Abstract

Upgrades Transolver with eidetic states and optimized parallel framework, demonstrating scalability to industrial-scale meshes with improved physics learning.

Usefulness

Rating: Moderately Useful

Scalability improvements relevant when extending to fine-mesh 2D problems, though eidetic states may be overkill for 1D Burgers.

Download

<https://arxiv.org/pdf/2502.02414>

23. Latent Generative Solvers

Abstract

Combines latent physics space with flow matching for generative temporal evolution, using on-manifold forcing to auto-correct trajectories and stabilize long rollouts.

Usefulness

Rating: Highly Useful

Generative approach with auto-correction could address your long-horizon stability; trained on 2.5M trajectories showing benefits of data scaling.

Download

<https://arxiv.org/pdf/2602.11229>

24. PhysicsCorrect: Training-Free Correction

Abstract

Projects predictions onto physically consistent manifolds via PDE residual Jacobian during inference, achieving 10-100x error reduction without retraining or architectural changes.

Usefulness

Rating: Highly Useful

Training-free correction could immediately improve your current TCAN; works with any pretrained model and maintains computational efficiency.

Download

<https://arxiv.org/pdf/2507.02227>

25. MMET: Multi-Input Multi-Scale Transformer

Abstract

Multi-scale Transformer handling multiple boundary conditions and geometries simultaneously, outperforming PinnsFormer and Transolver on complex multi-input datasets.

Usefulness

Rating: Moderately Useful

Multi-input encoding could help when you extend to variable boundary conditions in 2D Burgers and Navier-Stokes.

Download

<https://www.ijcai.org/proceedings/2025/0849.pdf>